

Response to Reviewers

Manuscript: “Edge Weight Concentration Overcomes Node Degree Blindness in Graph Based Network Intrusion Detection”

Discover Networks, Submission ID 5e70531c-36e0-427d-bc97-24cda1400b85

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We thank the editor and both reviewers for a genuinely constructive review. Reviewer 2 in particular identified two real methodological gaps – a cross-validation leakage risk and an untested causal claim – that materially changed what the manuscript can honestly claim, and we are grateful for the specificity of both reports. Below we respond to every point raised, in order, and indicate the corresponding change in the revised manuscript. Section and table numbers below refer to the revised manuscript.

Editor’s Internal Feedback

Comment: 1. Revise the title to read as one concise sentence without punctuation.

Response: Done. The title is now “Edge Weight Concentration Overcomes Node Degree Blindness in Graph Based Network Intrusion Detection” – a single declarative sentence with no colons, hyphens, or other structural punctuation.

Comment: 2. Note on comprehensiveness of referencing.

Response: Noted; no action required. We have, however, substantially expanded the Related Work section in direct response to Reviewer 2’s point 5 below, since those additions bear directly on the paper’s claims rather than being supplementary background.

Comment: Declarations: “Ethical approval”, “Consent to participate”, and “Consent to publish” as separate subheadings.

Response: Done. The Declarations section now presents these as three separate, individually labeled items rather than a combined “Ethics Approval and Consent to Participate” statement.

Reviewer 1

Comment: All empirical results come from a single CICIDS2017 DDoS capture containing one continuous attack event and one topology, which does not support broad generalization.

Response: We agree, and this remains the paper’s most important limitation (Section 7, item 1). We have not obtained additional captures for this revision. What we have done is strengthen what can honestly be concluded from the one capture available: Section 4.4 and 5.3 now report a controlled semi-synthetic sweep that varies the true collapse ratio k directly on this attack’s real flow volume, which tests the causal mechanism (identity collapse causes the degree/edge-weight divergence) rather than only observing it at one fixed ratio. This does not substitute for independent events or controlled NAT experiments across different topologies, which we now state explicitly as still-required future work (Section 7, item 2).

Comment: The flow-level evaluation may suffer from window-level information leakage because graph-context features are computed over time windows while individual flows are randomly assigned to cross-validation folds. Clarify feature-construction order and repeat the evaluation using window-grouped and temporally separated splits.

Response: This was a valid concern and we found a real effect when we tested it. Section 5.3 now reports classification performance under three protocols side by side (Table 3): the original naive StratifiedKFold on individual flows, GroupKFold grouped by window id (no window ever split across train/test), and a temporal split (train on the first 70% of windows chronologically, test on the last 30%). The baseline (78-dimensional CICFlowMeter) condition is essentially unaffected by protocol choice. The graph-context condition shows real degradation for Random Forest specifically (F1 falls from 1.000 naive to 0.981 grouped to 0.975 temporal), confirming the leakage risk was concentrated exactly where predicted – in the features that are constant within a window – and was larger for the higher-capacity Random Forest than for Logistic Regression. GroupKFold is now the primary reported protocol throughout the paper (Table 4), and the original naive numbers are retained alongside it rather than silently replaced, so the size of the effect remains visible to readers.

Comment: The 21 attack windows at 60-second resolution are temporally dependent observations from one event rather than independent attack instances, and the 600-second analysis contains only three attack and seven benign windows.

Response: We now state this explicitly in the evaluation-protocol description (Section 4.4) and repeat it at first use of the window-size table (Section 5.1): the four window sizes are nested aggregations of a single underlying one-minute-resolution timestamp series rather than independent replications of the experiment, and the 600-second condition rests on only ten windows total. We do not believe this invalidates the result – the effect is visible and significant at the finest, least-aggregated resolution (60 seconds, 93 windows) – but we agree the four window sizes should not be read as four independent confirmations, and the text no longer implies that they are.

Comment: Reporting inconsistencies, including $p < 10^{-4}$ in the abstract versus $p = 0.017$ at 600 seconds, should be corrected.

Response: Corrected. This was a genuine overstatement in the original abstract. The abstract now reports the true range across window sizes ($p = 3.9 \times 10^{-12}$ at 60 seconds to $p = 0.017$ at 600 seconds, the latter explicitly qualified as $n = 3$ attack windows) rather than a single blanket bound that did not hold at every resolution tested.

Comment: The claims should be narrowed: edge weight preserves aggregate traffic volume but does not recover the hidden number of attackers.

Response: We agree this is the more precise and defensible claim, and the collapse-ratio sweep now demonstrates it directly rather than asserting it: edge-weight separation is strong only near full collapse and does not recover once $k \geq 4$, showing edge weight tracks concentrated volume, not attacker count, exactly as the reviewer describes. Section 6.1 now states this explicitly: “edge weight preserves and reveals concentrated traffic volume on a host pair; it does not recover the number of real hosts that volume came from once they are collapsed – that quantity is genuinely destroyed by the collapse, not merely hidden.”

Comment: Additional independent events or controlled NAT experiments, corrected flow extraction, simple traffic-volume baselines, and edge-aware graph baselines would substantially strengthen the study.

Response: We addressed two of these four directly in this revision. *Traffic-volume baseline:* Section 5.1 now reports a volume-matched control comparing attack windows specifically against

the highest-volume benign windows in the capture, rather than all benign windows; edge-weight separation remains perfect against this harder comparison. *Edge-aware graph baseline*: Section 5.2 now reports weighted in-degree (node strength, the standard network-science combination of degree and edge weight), which also separates perfectly and is shown to be mathematically identical to max edge weight in this single-collapsed-pair setting – connecting our proposed feature to established network-science vocabulary rather than presenting it as an ad hoc quantity. *Independent events / controlled NAT experiments* and a full re-audit of flow extraction remain open; we did not have access to additional captures or a controlled testbed within the scope of this revision, and say so plainly in Section 7.

Reviewer 2

We are especially grateful for this report’s precision; both of the two core concerns below identify a genuine gap between what the theory claims and what the original experiment measured, and both admitted a direct, testable fix, which we implemented.

Comment: 1. The experiment does not vary the quantity the argument is about (k). A controlled version would progressively rewrite several visible source addresses onto one and plot both separations against the collapse ratio.

Response: We implemented exactly the experiment described. Section 4.4 describes the design and Section 5.2 reports the result (Figure 3, Table 2): holding the real attack’s timestamps, victim address, and behavioral features fixed, we redistribute its flow volume across $k \in \{1, 2, 4, 8, 16, 32, 64, 128, 256, 1024, k_{\max}\}$ synthetic source identities and recompute both separation curves. The result matches the theoretical prediction in its qualitative shape (degree recovers separation as k grows; edge weight is strong only near full collapse) but also surfaces something the closed-form argument alone did not anticipate: a partial-collapse regime ($4 \lesssim k \lesssim 16$) in which *neither* feature family separates well, with edge-weight AUC dropping to exactly chance level (0.50) at $k = 4$ before plateauing around 0.61. We report this as a genuine open finding rather than a clean confirmation, and flag in Section 7 that this sweep is semi-synthetic (uniform-random redistribution on one real attack’s volume), not independent real events at differing collapse ratios, so it strengthens but does not fully replace the validation the reviewer is asking for. On the second part of this comment: we could not find independent evidence beyond the topology match already reported (Section 3) to substantiate that the two observed addresses reflect true NAT translation rather than a small attacker set observed as-is; we now state this as an assumption consistent with, but not proven by, the available evidence, rather than asserting it.

Comment: 2. The feature and the window label are nearly the same quantity, since the attack sits entirely on one pair. This near-tautology deserves its own sentence, plus a volume-matched control. The same issue affects the flow-level ablation, where two of the ten features are close to the label; reporting the graph-context condition without the heaviest-edge indicator would make the efficiency claim cleaner.

Response: We addressed both parts. *Quantifying the near-tautology*: Section 5.1 now reports the direct correlation between attack flow fraction (the basis of the window label) and edge-weight top-1 share across all 93 windows: Pearson $r = 0.949$, Spearman $\rho = 0.732$ ($p = 7.6 \times 10^{-17}$), stated plainly as evidence the window-level AUC is, in part, a near-restatement of the labeling rule. *Volume-matched control*: also in Section 5.1, we repeat the test restricting the benign comparison set to the 21 highest-volume benign windows; edge-weight separation remains perfect (AUC = 1.00, $p = 3.1 \times 10^{-8}$) against this harder control. *Cleaner efficiency claim*: we removed the heaviest-

edge binary indicator from the graph-context feature set used in the primary classification result (Section 5.3, Table 4), leaving nine features. Under this cleaned set and GroupKFold evaluation, the efficiency claim changes from “matches baseline accuracy” to a more modest and more honest one: $F1 = 0.981$ against the baseline’s 0.998–1.000, at roughly one-tenth the training cost – a real accuracy cost in exchange for the dimensionality and cost reduction, not a free win. The original ten-feature ablation numbers (Table 5) are retained as reported, with an explicit note that the “all 10” row includes the indicator later removed from the primary result, so nothing is silently changed without a visible trace.

Comment: 3. Independence of the window-size check. 60s is the atomic bin given one-minute timestamp resolution, so 120/300/600s are nested aggregations, not independent samples; the abstract’s “across four window sizes” reads as four confirmations. At 600s, $AUC=1.00$ on $n=3$ vs. 7 would arise by chance fairly often.

Response: Addressed directly (this overlaps with Reviewer 1’s related point above). The evaluation-protocol text (Section 4.4) and the window-level results text (Section 5.1) now state explicitly that the four window sizes are nested aggregations of one underlying series, and the abstract’s phrasing no longer implies four independent confirmations. We do not remove the 600-second result, since it is still the correct Mann-Whitney p -value for that (small) sample and we would rather show it honestly caveated than omit it, but we agree the reader should not weight it as equal evidence to the 60-second result, and the text now says so.

Comment: 4. Evadability of edge-weight concentration. An attacker aware the detector keys on concentration can spread volume across many ports or victims, restoring entropy cheaply – the mirror image of degree evasion.

Response: Added. Section 6.1 now includes a paragraph titled “Identity-robustness is not evasion-robustness,” stating this directly: edge-weight concentration and node degree are each robust to the topological condition the other is blind to, but neither is evasion-robust against an attacker who specifically targets it, and we did not evaluate adversarial evasion in this study (now also listed as an explicit limitation, Section 7, item 4). We point to the frequency-domain (Whisper) and causal-analysis (Wedjat) approaches surveyed in the expanded Related Work as designed specifically for evasion robustness, which this paper’s feature family is not.

Comment: 5. Expand the Related Work beyond graph-NIDS architectures: HyperVision, Whisper, Wedjat, nPrintML, packet-length semantic analysis, tFusion, FlowLens, NetBeacon, pVoxel.

Response: All ten references have been incorporated across two new Related Work paragraphs (Section 2): “Traffic-analysis systems and what graph representations should encode” (HyperVision, Whisper, Wedjat, nPrintML, packet-length semantic analysis, tFusion) and “Deployment and data-plane systems” (FlowLens, NetBeacon, pVoxel), engaging each on the specific point the reviewer raised rather than listing them as background – in particular, HyperVision as the closest existing system to our framing that does not examine identity collapse, Whisper/Wedjat as connecting to the evasion-robustness discussion above, and FlowLens/NetBeacon/pVoxel as establishing the deployment feasibility ceiling and the false-positive-volume concern that Section 6.4’s efficiency argument should be read against.

Summary of Substantive Changes

- New controlled collapse-ratio sweep (Section 4.4, 5.2, Figure 3, Table 2), directly testing the causal identity-collapse claim rather than inferring it from one fixed ratio, and revealing a previ-

ously unreported partial-collapse regime where neither feature family separates well.

- Cross-validation leakage identified and fixed: GroupKFold by window id is now the primary evaluation protocol, with the original naive-split numbers retained alongside it for transparency (Section 5.3, Tables 3–4).
- Near-tautological heaviest-edge indicator removed from the primary graph-context feature set; efficiency claim revised from “matches baseline” to a quantified accuracy/efficiency trade-off.
- Near-tautology between the edge-weight feature and the window label quantified directly ($r = 0.949$) rather than left implicit.
- Volume-matched benign control added (Section 5.1).
- Edge-aware weighted-degree (strength) baseline added, connecting the paper’s central feature to standard network-science vocabulary (Section 5.2).
- Evasion-robustness discussion added, distinguishing it explicitly from the identity-robustness this paper studies (Section 6.1).
- Abstract’s $p < 10^{-4}$ overclaim corrected to report the true range.
- Related Work expanded with ten additional references spanning traffic-analysis representation systems and deployment/data-plane systems.
- Title revised to a single punctuation-free sentence; Declarations restructured into the three requested separate subheadings.

We believe these changes leave the paper’s central claim – that identity collapse under NAT is a real, previously unaddressed failure mode for node-centric graph-NIDS features, and that edge-weight concentration is a partial, mechanistically-understood, but not universal remedy – more precisely stated and more directly tested than in the original submission, and we have tried throughout to make each remaining limitation as visible as the results themselves.